

Nicholas J. Calabro
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Education:

B.S. in Computer Science

University of Massachusetts Lowell, Lowell, MA

Expected May 2026

Work Experience:

Olympic Pool Service, Quincy MA: Pool Technician (May 2021 – November 2023):

- Completed dozens of openings, closing, and liner replacements working with a small crew.
- Performed an independent pool service route of 40 consistent properties a week.
- Ensured pool uptime by adjusting chemicals and equipment to health department standards.

Cogo Labs, Cambridge MA: Go Software Engineer Intern (May 2020 - August 2020):

- Designed, implemented, deployed, and monitored two projects written in Go and TypeScript, each ran in production Kubernetes containers deployed through Travis CI.
- Developed both projects using Git while working with and learning from a team.
- Planned and managed project progression using GitHub issues to facilitate communication.
- Requested and analyzed code reviews from other engineers, resulting in new improvements.
- Implemented comprehensive go test suites with broad code coverage, improving code quality.

DMARC Reporter:

- Protected key revenue streams by implementing a DMARC reporter to ensure email deliverability.
- Queried an Elasticsearch Engine for thousands of daily DMARC report emails from providers.
- Parsed the reports, encoded in zipped base64 into raw XML, then inserted the XML report data into a Postgres database table, providing a PostgreSQL interface for company engineers.
- Created a robust monitoring service with Prometheus and Sysdig to graph email failure rates, system status, and performance.
- In the event of significant send failures, providing these interfaces will eliminate or flag DMARC, either DKIM or SPF, as point of failure quickly reducing response times and lost revenue.

Invoice Scraper:

- Parsed submitted payments from invoice PDFs for leased IP addresses, automating a previously manual sysadmin task.
- Developed an Angular web application to submit invoices and to display the parsed results.
- Engineered a YAML-based configuration system for the invoice scraper backend, which facilitated the parsing of invoices from at least 5 different providers with varying syntax.
- Prepared intuitive markdown documentation about creating or changing configurations to minimize labor time if the invoice syntax changed or a new IP provider required parsing.
- Incorporated optional output to display how the parser executes to indicate configuration errors, decreasing required manual time to configure the parser.

Skills:

Languages: Go, C++, C, Java, TypeScript, Python, HTML, CSS, SQL

Frameworks, Tools, Technologies: Git, Docker, Kubernetes on AWS, Angular, Vim

Databases, Search Engines: PostgreSQL, Oracle Database, Elasticsearch

High-level Classes:

Computing (with AI and ML) for Health and Medicine, Introduction to Linux Development, Database I and II, Operating Systems, Analysis of Algorithms, Computer Architecture, Assembly Language